

Problem Chosen

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Summary Sheet

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Your Paper's Title(SHUMOJIAYOUZHAN)

Summary

hellow everyone

Keywords: keyword1; keyword2

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1 Introduction

In professional sports, a long-standing tension exists between the pursuit of athletic glory and the necessity of financial sustainability. While fans focus on the "tip of the iceberg"—the players on the court—owners must manage the complex business machinery beneath. For leagues like the Women's National Basketball Association (WNBA), this challenge is magnified as they transition from risky startups to major entertainment enterprises.

Our objective is to design a dynamic decision-making model that helps owners balance these competing goals. We focus on the following two core dimensions of value:

- **On-Court Performance Index (*OPI*):** Measuring a player's direct contribution to winning games through advanced statistics.
- **Commercial Value Index (*CVI*):** Quantifying a player's ability to generate revenue via ticket sales, jersey sales, and media engagement.

The complexity of modern sports management stems from several volatile factors:

- The non-linear relationship between winning games and generating profit.
- The temporal element of player value, including future potential and injury risks.
- Market dynamics influenced by team location and league-wide constraints like salary caps.

1.1 Our Approach

In this paper, we propose a multi-objective optimization framework to guide team management. By integrating *OPI* and *CVI*, we provide a roadmap for roster construction and financial leveraging.

Theorem 1.1. *Optimal Roster Efficiency occurs when the marginal cost of *OPI* equals the marginal revenue generated by *CVI*, subject to the league salary cap.*

1.2 Core Assumptions

To ensure the model is both robust and applicable, we establish the following assumptions:

- **Rational Management:** Owners aim to maximize the combined utility of team profit and asset value.
- **Data Availability:** Publicly available sports and financial data are treated as accurate representations of past performance.

- **Systemic Constraints:** League rules, such as salary caps and roster limits, remain constant unless an expansion is explicitly modeled.
- **Market Liquidity:** Players can be traded or acquired through standard practices like free agency or drafts.

2 Preparation of the Models

2.1 Assumptions

To simplify the complex environment of professional sports and ensure the mathematical tractability of our model, we establish the following core assumptions:

- **Discrete Decision Epochs:** Management evaluates team status and makes strategic adjustments at discrete intervals (e.g., monthly or every 10 games), rather than continuously.
- **Decoupling of Performance and Commercial Value:** A player's total value can be decomposed into independent athletic (OPI) and commercial (CVI) components, which are not always positively correlated.
- **Rational Financial Management:** The team owner acts as a rational economic agent aiming to maximize profit and asset value while strictly adhering to debt constraints such as the Debt Service Coverage Ratio (DSCR).
- **Non-linear Impact of Key Injuries:** The injury of a "key player" has a disproportionate, non-linear downward impact on the team's health index H_t , triggering an immediate defensive shift in management strategy.
- **Price Elasticity of Demand:** Market demand for tickets and merchandise is sensitive to pricing and team performance, following an exponential decay function relative to price increases.
- **Structural Expansion Shocks:** League expansion is treated as an external shock that impacts existing teams through talent dilution, revenue redistribution, and increased travel burdens.

2.2 Notations

To provide a clear structure for our mathematical framework, we categorize the primary symbols into four functional groups: General State, Financial Management, Player Valuation, and Business Strategy.

Table 1: Summary of Notations Categorized by Modeling Tasks

Symbol	Description	Unit/Range
Group 1: General State & Performance (Q1 & Q3)		
t	Decision step or epoch (e.g., monthly or 10-game block)	-
ES_t	Environmental Score (composite signal for state transition)	$[0, 1]$
OPI_t	On-court Performance Index (athletic efficiency metric)	$[0, 1]$
CVI_t	Commercial Value Index (brand asset and monetization potential)	$[0, 1]$
W_t, H_t	Normalized winning percentage and Team Health index	$[0, 1]$
r_t	Prevailing market financing/interest rate index	$[0, 1]$
Group 2: Financial Management & Leverage (Q1)		
L_t	Leverage ratio (Debt-to-Equity ratio)	Ratio
D_{max}	Debt ceiling (maximum sustainable debt based on $DSCR$)	Currency
$EBITDA_t$	Earnings before interest, taxes, depreciation, and amortization	Currency
$DSCR$	Debt Service Coverage Ratio (minimum threshold for solvency)	Ratio
Group 3: Player Valuation & Acquisition (Q2)		
V_i^{sport}	Athletic value of player i (based on performance and potential)	Scalar
V_i^{biz}	Commercial value of player i (based on brand and marketability)	Scalar
x_i	Binary decision variable ($x_i = 1$ if player i is acquired/retained)	$\{0, 1\}$
$\lambda(ES)$	State-driven weighting factor for "Winning vs. Monetization"	$[0, 1]$
Group 4: Business Strategy - Ticket Pricing (Additional Task)		
p_g	Price of a single ticket for game g	Currency
$D_g(p)$	Market demand function for game g at price p	Scalar
A_g	Base demand multiplier (reflecting rivalry and star power)	Scalar
k	Price sensitivity (elasticity) coefficient	Constant

3 The Dual-Index Model

Professional sports teams must balance two distinct but interconnected objectives: on-court success and commercial profitability. Winning games drives fan engagement and revenue, yet popularity, market size, and macroeconomic conditions can generate substantial value even in periods of mediocre performance. To capture this duality, we propose a unified decision-making framework centered on two independent yet linkable indices—the On-court Performance Index (OPI) and the Commercial Value Index (CVI)—combined with a dynamic state signal, the Environmental Score (ES), that drives conditional strategy selection through a three-state regime.

3.1 The Dual-Index Framework

3.1.1 On-court Performance Index (OPI)

The On-court Performance Index compresses competitive strength into a single normalized metric in $[0, 1]$:

$$OPI_t = \omega_1 \cdot WinRate_t + \omega_2 \cdot NetRating_t + \omega_3 \cdot PlayoffProb_t,$$

where weights $\omega_1 + \omega_2 + \omega_3 = 1$ (default: $\omega_1 = 0.5, \omega_2 = 0.3, \omega_3 = 0.2$). When detailed data are unavailable, OPI may be approximated by normalized win rate or Elo rating alone. OPI measures “winning ability” but does not necessarily correlate perfectly with financial returns.

3.1.2 Commercial Value Index (CVI)

The Commercial Value Index quantifies the team’s ability to convert performance and brand appeal into revenue and long-term asset value:

$$CVI_t = \nu_1 \cdot Revenue_t + \nu_2 \cdot Engagement_t + \nu_3 \cdot BrandValue_t,$$

with $\nu_1 + \nu_2 + \nu_3 = 1$ (suggested: $\nu_1 = 0.4, \nu_2 = 0.3, \nu_3 = 0.3$). Revenue includes ticket sales, media rights shares, sponsorships, and merchandise; Engagement encompasses attendance rates and social media interactions; BrandValue is proxied by jersey sales rankings, All-Star votes, and market-adjusted follower growth. CVI explains why teams with high popularity but moderate records can remain financially attractive to owners.

3.2 Environmental Score (ES) as State Signal

Rather than optimizing a single static objective function, we adopt a state-driven conditional optimality approach: strategies adapt based on prevailing conditions. To operationalize this, we define a composite Environmental Score $ES_t \in \mathbb{R}$ that integrates competitive, operational, and macroeconomic signals:

$$ES_t = \alpha W_t + \beta H_t - \gamma r_t, \quad \alpha + \beta + \gamma = 1,$$

where

- $W_t \in [0, 1]$: normalized recent win rate,
- $H_t \in [0, 1]$: health/availability index (weighted attendance rate of core rotation players),
- $r_t \in [0, 1]$: normalized financing cost (e.g., federal funds rate or 10-year Treasury yield proxy).

Default weights are $\alpha = 0.4, \beta = 0.3, \gamma = 0.3$. High win rates and player health improve investment attractiveness, while rising interest rates increase the cost of leverage.

ES Range	State	Target Leverage
$ES_t > 0.6$	Aggressive	High (≈ 1.0)
$0.3 \leq ES_t \leq 0.6$	Neutral	Moderate (≈ 0.5)
$ES_t < 0.3$	Defensive	Low (≈ 0)

Table 2: Three-state regime driven by Environmental Score.

3.3 Three-State Regime and Action Mapping

The value of ES_t determines the operating state and corresponding strategic posture:

- **Aggressive State** ($ES_t > 0.6$): Favor growth and championship pursuit. Target high leverage to finance star acquisitions, premium ticket pricing, and aggressive marketing.
- **Neutral State** ($0.3 \leq ES_t \leq 0.6$): Balance growth and stability. Maintain moderate leverage, pursue cost-effective roster improvements, and optimize operations.
- **Defensive State** ($ES_t < 0.3$): Prioritize cash-flow safety and risk reduction. Delever aggressively, shed high-risk contracts, and focus on attendance retention through stable pricing.

This state machine enables conditional optimality: in favorable environments the framework favors growth and championship pursuit; in adverse conditions it prioritizes cash-flow safety and risk reduction. Safety constraints (Debt Ceiling and Equity Trigger, detailed in Section 4) prevent excessive risk even in Aggressive states.

The unified Dual-Index Model with ES-driven states serves as the core engine for all subsequent analyses: dynamic leverage control (Q1), player acquisition weighting (Q2), expansion impact assessment (Q3), ticket pricing policy, and automatic injury response.

4 Model I: Dynamic Leverage Control Model

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5 Sensitivity Analysis

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6 Model Evaluation and Further Discussion

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6.1 Strengths

- **Applies widely**

This system can be used for many types of airplanes, and it also solves the interference during the procedure of the boarding airplane, as described above we can get to the optimization boarding time. We also know that all the service is automate.

- **Improve the quality of the airport service**

Balancing the cost of the cost and the benefit, it will bring in more convenient for airport and passengers. It also saves many human resources for the airline.

-

7 Conclusion

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References

- [1] D. E. KNUTH The T_EXbook the American Mathematical Society and Addison-Wesley Publishing Company , 1984-1986.
- [2] Lamport, Leslie, L^AT_EX: " A Document Preparation System ", Addison-Wesley Publishing Company, 1986.
- [3] <http://www.latexstudio.net/>

[4] <http://www.chinatex.org/>

Appendices

Appendix A First appendix

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Input matlab source:

```
function [t,seat,aisle]=OI6Sim(n,target,seated)
pab=rand(1,n);
for i=1:n
    if pab(i)<0.4
        aisleTime(i)=0;
    else
        aisleTime(i)=trirnd(3.2,7.1,38.7);
    end
end
```

Appendix B Second appendix

some more text **Input C++ source:**

```
//=====
// Name      : Sudoku.cpp
// Author     : wzlf11
// Version    : a.0
// Copyright  : Your copyright notice
// Description : Sudoku in C++.
//=====

#include <iostream>
#include <cstdlib>
#include <ctime>

using namespace std;

int table[9][9];

int main() {

    for(int i = 0; i < 9; i++){
```

```
        table[0][i] = i + 1;
    }

    srand((unsigned int)time(NULL));

    shuffle((int *)&table[0], 9);

    while(!put_line(1))
    {
        shuffle((int *)&table[0], 9);
    }

    for(int x = 0; x < 9; x++){
        for(int y = 0; y < 9; y++){
            cout << table[x][y] << " ";
        }

        cout << endl;
    }

    return 0;
}
```
